SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	**Date:**	28-08-2026

Code of Conduct

I ____T Bhanu Teja/192472259_____________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____BhanuTeja_________

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform Reference Resolution by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

ANS:

Algorithm

Input: A discourse containing pronouns
Output: Discourse with pronouns replaced by their correct entities

Algorithm Reference_Resolution(Text)

1. Read the input text.

2. Split the text into sentences.

3. Identify all entities (nouns/names) in the text.

4. Identify pronouns such as He, She, It, They, etc.

5. For each pronoun:

    a. Find possible antecedents from previous sentences.

    b. Check gender and number agreement.

    c. Check grammatical and semantic compatibility.

    d. Use discourse context to select the most suitable antecedent.

6. Replace each pronoun with its selected antecedent.

7. Return the resolved discourse.

End Algorithm

Application

Original:

"Ravi met Arun at the library. He borrowed a book and later returned it."

- He → Ravi because Ravi is the most suitable antecedent in the context.
- it → book because a book is an object that can be returned.

Resolved Discourse

"Ravi met Arun at the library. Ravi borrowed a book and later returned the book."

Validation

The algorithm uses discourse context, previous nouns, grammatical agreement, and semantic meaning. The pronoun "He" refers to Ravi, while "it" refers to book. This makes the discourse clear and meaningful**.**

2. Develop an algorithm to analyze Text Coherence and Discourse Structure by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

ANS:

Algorithm

Algorithm Analyze_Coherence(Text)

1. Read the input discourse.

2. Split the discourse into sentences.

3. Identify important events or facts in each sentence.

4. Compare neighboring sentences.

5. Identify logical relationships:

    a. Cause-Effect

    b. Sequence

    c. Elaboration

    d. Contrast

6. Create a discourse structure connecting the sentences.

7. Return the identified relations.

End Algorithm

Application

Discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Discourse Relations

1. **Sentence 1 → Sentence 2: Cause-Effect**
   - Heavy rainfall caused flooding.
   - Flooded roads resulted in schools being closed.
2. **Sentence 2 → Sentence 3: Sequence/Result**
   - Schools were closed.
   - As a result, students attended classes online.

Discourse Structure

Heavy Rainfall

↓

Roads Flooded

↓

Schools Closed

↓

Students Attend Online Classes

Representation

S1: Roads were flooded.

|

CAUSE

↓

S2: Schools were closed.

|

RESULT

↓

S3: Students attended online classes.

Justification

These relations maintain **coherence** because each sentence is logically connected to the next. The flooding explains the school closure, and the school closure explains why students attended classes online.

3. Design an algorithm for a Conversational Agent that identifies Dialogue Acts from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

ANS:

Algorithm

Algorithm Identify_Dialogue_Act(Utterance)

1. Read the user's or agent's utterance.

2. Check for an explicit request.

3. If it asks for something → classify as REQUEST.

4. If it asks for information → classify as QUESTION.

5. If it provides information → classify as INFORM.

6. If it confirms an action → classify as CONFIRMATION.

7. If it performs or indicates an action → classify as ACTION.

8. Return the dialogue act.

End Algorithm

Application

Conversation:

Speaker	Utterance	Dialogue Act
User	"Can you book a train ticket for me?"	**Request**
Agent	"Sure, where would you like to travel?"	**Question**
User	"I want to go to Chennai."	**Inform**
Agent	"Your ticket has been booked."	**Confirmation**

Dialogue-Act Sequence

REQUEST → QUESTION → INFORM → CONFIRMATION

Evaluation

Dialogue acts help the conversational agent understand the user's intention.

- **Request** tells the agent what the user wants.
- **Question** collects missing information.
- **Inform** provides the required information.
- **Confirmation** tells the user that the requested action has been completed.

Thus, dialogue-act identification helps the agent conduct the conversation correctly.

4. Design an algorithm for Language Generation that converts a semantic representation into a grammatically correct sentence through Surface Realization. Consider the semantic input:

```
Action: Buy
Agent: Student
Object: Book
Tense: Past
```

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

ANS:

Semantic Input

Action : Buy

Agent : Student

Object : Book

Tense : Past

Algorithm

Algorithm Surface_Realization(Semantic_Input)

1. Read the semantic representation.

2. Identify the main action.

3. Select appropriate words from the vocabulary.

4. Identify the agent and object.

5. Arrange them according to English sentence structure:

   Subject + Verb + Object

6. Convert the verb into the required tense.

7. Add necessary articles and grammatical elements.

8. Check grammar and agreement.

9. Generate the final sentence.

End Algorithm

Steps

1. Lexical Selection

Agent → Student

Action → Buy

Object → Book

2. Sentence Structure

Subject + Verb + Object

Student + bought + a book

3. Tense Conversion

Buy → Bought

4. Surface Realization

"The student bought a book."

Validation

The generated sentence is grammatically correct because:

- **"The student"** acts as the subject.
- **"bought"** is the past tense of *buy*.
- **"a book"** is the object.
- The sentence follows **Subject–Verb–Object (SVO)** structure.

Therefore, the final output is:

The student bought a book.

5. Design an algorithm that combines Interlingua-based Machine Translation and Statistical Translation Approaches. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. **Analyze the source sentence.**
2. **Create an Interlingua representation.**
3. **Generate candidate target-language translations.**
4. **Apply statistical scoring to select the most appropriate translation.**
5. **Produce the final translated sentence.**

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

ANS:

Algorithm

Algorithm Hybrid_Machine_Translation(Source_Sentence)

1. Read the source sentence.

2. Perform lexical and syntactic analysis.

3. Identify the meaning of the sentence.

4. Convert the meaning into an Interlingua representation.

5. Generate possible target-language translations.

6. Calculate statistical scores for each candidate.

7. Select the candidate with the highest score.

8. Perform grammatical correction.

9. Produce the final translation.

End Algorithm

Given Sentence

"The boy is playing football."

Step 1: Analyze Source Sentence

Identify the components:

Subject → The boy

Action → is playing

Object → football

Tense → Present Continuous

Step 2: Interlingua Representation

The language-independent meaning can be represented as:

PLAY(

AGENT = BOY,

OBJECT = FOOTBALL,

TENSE = PRESENT_CONTINUOUS

)

This represents the meaning without depending on a particular language.

Step 3: Generate Candidate Translations

For example, for **Tamil**, possible candidates could be:

Candidate 1: சிறுவன் கால்பந்து விளையாடுகிறான்.

Candidate 2: சிறுவன் கால்பந்து விளையாடுகின்றான்.

Step 4: Statistical Scoring

The system evaluates candidates using factors such as:

Translation Probability

+

Language Model Probability

+

Grammar

+

Context

Example:

Candidate 1 → Score = 0.92

Candidate 2 → Score = 0.84

Therefore, Candidate 1 receives the higher score.

Evaluation

The Interlingua representation captures the core meaning of the source sentence independently of language. This reduces ambiguity during translation.

The statistical approach compares possible translations and selects the one that is more probable and natural in the target language.

Therefore, combining both approaches improves translation accuracy, grammatical quality, contextual understanding, and ambiguity reduction.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________